

Design of experiment for evaluation of anomalous environments for product visualization in virtual reality

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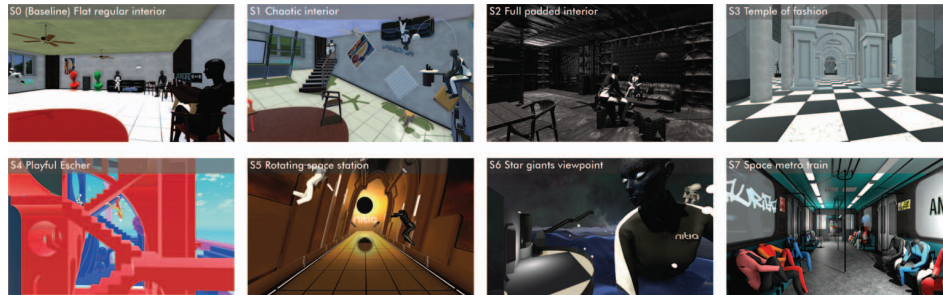


Figure 1: Overview on all scenes designed for the experimentation: on top left the baseline scene

ABSTRACT

Virtual reality metaverse is candidate to be the future platform for retail. Nevertheless, the knowledge in designing virtual environments and their influence on products is limited in literature. Common “brick and mortar” virtualization of shops as realistic digital twin is a reductive approach compared to more creative experiences. This work explores novel paradigms by evaluating “anomalous environments” deviated from the ordinary conception. Different anomalies are evaluated within 8 scenes promoting an existing line of futuristic-looking tracksuits. The experimentation on 42 subjects, shows that anomalous environments, especially when aligned with the product’s storytelling, can increase products likeability, interest and perceived valorization.

Keywords: Metaverse, retail, fashion, perception, storytelling, product valorization, virtual environments, virtual reality.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods—; Human-centered computing—Visualization—Visualization theory, concepts and paradigms

1 INTRODUCTION

This research explores the potential of virtual reality (VR) to revolutionize the retail landscape, exploring its ability to create immersive environments, connect the physical and digital realms, and facilitate data exchange [5]. With research and development in this area poised to surge, the impact on branding, service offerings, and supply chain sustainability is wide. Interaction and perception [4] play a crucial role in this new frontier, and the fashion industry serves as an ideal case study, having faced the use of VR for virtual showrooms, stores, and promotional virtual events. While traditional approaches often replicate physical stores in VR, this research proposes an alternative: anomalous environments that intentionally deviate from reality.

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These environments introduce incongruities based on plausibility and expectations, stimulating sensory, spatial, logical, and cognitive engagement such as size, lack, repetition and placement [2] [3]. Specific parameters, such as dynamism (the conscious choice to make objects and architecture movable and not static) and toon shaders (fanciful and fictional visualization style in contrast to the realistic one), are added in order to enhance the anomalous effect. The research’s objective is to verify if anomalous environments that better leverage virtual reality’s ability to go beyond the ordinary are more effective in both appreciating the product and the technology itself. The expected impact is to encourage the design of less conservative experiences, also related to the retail context, where greater appreciation leads to higher economic returns, as well as to encourage creative experimentation with these new media.

2 RESEARCH QUESTIONS

- 1. Are anomalous environments more capable in valorizing products?
- 2. Does a major attention in environment affect negatively the product interest?

3 METHODOLOGY

Upon detecting anomalies in the parameter settings of the scenes to be created, two questionnaires were developed. The first questionnaire, administered prior to the experiment, after sought the subject’s consent for anonymous data collection for research purposes, encompassed inquiries regarding gender, age, occupation, any prior experiences with VR and familiarity with video games expressed in gaming hours per week. The second questionnaire, administered post-exploration of the virtual scene within each scenario for one minute of silence, was designed by incorporating elements from various metrics. Specifically, participants were prompted to rate, on a Likert scale, the following statements:

- 1. “I felt present in the virtual space”- strongly disagree–strongly agree (IPQ) [1]
- 2. “Somehow I felt that the virtual world surrounded me” - strongly disagree–strongly agree (IPQ) [1]
- 3. “Does the environment appear anomalous?” - not strange/weird/abnormal–strange/weird/abnormal [3]

- 4. "Do you perceive the environment as realistic?" - not real/authentic – real/authentic [3]
- 5. "Do you find the scenery surrounding you familiar?" - new/stranger/unfamiliar – known/regular/familiar [3]

Then, since no prior scientific literature was identified regarding the relationship between the environment's design and its influence, but only non-academic speculation (keywords used for research: product, environments, scenarios, effects, perception, store, interior design, valorization, scene) tailored questions were formulated:

- 6. "How do the products appear within the environment?" - "not at all appealing–very appealing"
- 7. "Does the product manage to surpass the environment in importance?" - the environment – the product
- 8. "Do you believe the environment valorize the products?" - definitely not – definitely yes

The administration of this second questionnaire took place between one scene and another before the next environment was loaded by entering the corresponding number on the keyboard thanks to a scene loading script. Following surveys, mood boards were created to assist in designing 8 virtual environments within Unity. The futuristic suits Andromeda, Cassiopea, Moon, and Auriga, from the Nitia Labtech line, were provided by Emmeevolution in Martina Franca (TA) for their 3D models. Additional 3D assets were either created using Rhinoceros 3D or downloaded under Creative Commons license. The Unity application employed Open XR Toolkit and was tested on Oculus Rift and Oculus Quest 2. The final experiment was conducted on Oculus Rift in a university laboratory. The average experiment duration was 25 minutes, and each scene averaged 4 minutes. The final scenes (Fig.1) have been designed and developed in this way: Scene 0, baseline, "Flat regular interior": Realistic proportions and materials, realistic architecture, realistic shader, static environment; Scene 1, "Chaotic interior": asymmetry, unrealistic furniture disposition, unrealistic architecture, mixed shaders, static environment, realistic product size; Scene 2, "Full padded interior": monochrome, mono-material, unrealistic architecture, realistic shader, static environment, realistic product size; Scene 3, "Temple of fashion": symmetry, massive furniture, repetition, realistic architecture, realistic shader, static environment, massive product presentation; Scene 4, "Playful Escher": massive environment, vertical deploy, unrealistic architecture, monochrome, toon shader, static, environment, realistic product size; Scene 5, "Rotating space station": massive environment, unrealistic architecture, realistic shader, dynamic environment, realistic product size; Scene 6, "Star giants viewpoint": massive product, unrealistic, architecture, realistic shader, static, environment, massive product size; Scene 7, "Space metro train": realistic proportions and material, repetition, realistic architecture, dynamic environment, realistic product size. Data collection transpired via a Google Form, with a research team member verbally posing the questions and documenting the responses each time.

4 COLLECTED DATA

Of the 42 subjects from the university environment: 22 were males, 19 females, 1 other gender. The average age was 27 years old ($\pm 7,5$ SD). 32 of the subjects that made the experiment already had a previous VR experience, while 10 were first timers. The average gaming hours per week was 2,8 (5,1 SD). The first data about the immersivity show a really positive capability of each scene to communicate and being consistent, with a general average of 6,44 (1,16 SD) in feeling inside the environment and a score of 6,10 (1,43 SD) in perceiving the scene around. All the data collected for each scene (s1, s2...) has been compared with the feedback about the first baseline scene (fig 2.)

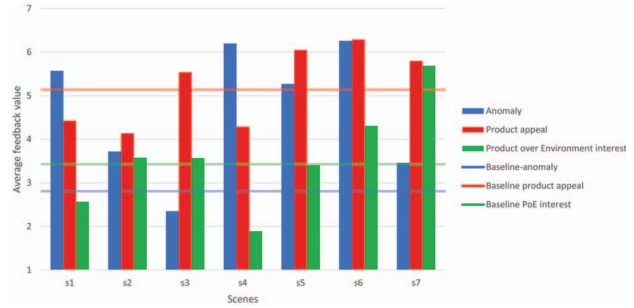


Figure 2: Valutative parameters comparison graphic

5 DISCUSSION

Hypothetically, anomalous environments, given their unusual and unexpected nature, are able to engage the user's curiosity more, consequently increasing his or her attention and pleasure in discovering what is around them. The concept of novelty therefore should be predictive of a greater appreciation of each element of the environment precisely because of the paradigm shift. From the data collected and in particular the comparative graph (fig. 2), it appears that this at the virtual reality level, is not a foregone conclusion. In fact, the anomalous environments affected positively compared to the baseline with the product when their concept remained consistent and coherent with the storytelling of the product (s5, s6, s7). In addition, environments such as s4 where the environment was perceived as highly anomalous, the user's attention to the product was diverted. As explanation, even though younger people, main subjects of the experiment, are typically more receptive to new things, the traditional showroom experience remains significant for the fashion industry where subjective appreciation plays a crucial role.

6 CONCLUSIONS

The tendency in the industry to resort to conservative concepts is due to ease of familiar concepts to meet an average product approval, however when the scenario is designed "anomalous" but with a definite and consistent storytelling purpose, within this experiment this has proven capable of significantly increasing the appreciation. What has emerged from subject's feedback is that the major challenge in designing disruptive virtual environments aimed to retail is to miss the point of keeping the importance of product as central element. For further research works, will be also important to consider the subjects pool in terms of taste, especially in fashion, and background, alongside reproduce experiments with different products that require indeed different approaches.

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